**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# LESSON 223 · Linked lists using objects

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What makes a heuristic admissible?

Q2. What does a partition do to the pivot?

Q3. What does the head reference identify?

Q4. Why must a new node's Next be set before replacing an existing Next link?

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

node — An object containing data and a reference to another node.

head — Reference to the first node.

Next — Link to the successor node.

null — No object is referenced.

traversal — Visit nodes successively by following Next references.

## New learning

Build a chain — Each Node stores a Value and a Next reference; the final Next is null.

Preserve the successor — Construct 15 with Next set to the old 20 node *before* changing 10.Next.

Finish the insertion — Set 10.Next to 15. Now traversal yields 10→15→20→30.

## Worked example

Insert node 7 between 4 and 9 in 4 → 9 → 12.

1. Predict first — Set newNode.Next to the existing successor 9.

2. Model the reasoning — Update node4.Next to reference the new 7 node. The 9→12 remainder is preserved.

# We Do · Practise with guidance

Insert 6 after node 2 in 2 → 8 → 11.

```
Node newNode = new(6, node2.Next);
node2.Next = newNode;
```

Predict first

First give the new node a reference to 8, the former successor of 2.

Model the reasoning

Then change node2.Next to reference 6; the final chain is 2→6→8→11.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

For 3→5→8, identify head, 5.Next.Value and 8.Next.

### Question 2

Write two C# statements to insert a node with value 12 after node8 in 8→20→30. Give the new traversal.

### Question 3

Describe how to delete node 20 from 10→20→30 if you hold a reference to node10. Explain the complexity of reaching the deletion point and changing the link.

## Exit Ticket · Answer before checking

Inserting a linked-list node always requires shifting the stored values of later nodes.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. It never overestimates the true cheapest remaining cost.

Q2. Places it at its final index with smaller/equal values to its left.

Q3. The first node in a linked list.

Q4. Otherwise the original successor can become unreachable and part of the list may be lost.

### You Do pages · Independent questions

Q1. [3] Head is the node containing 3 [1]; 5.Next.Value is 8 [1]; 8.Next is null [1].

Q2. [5] Node inserted = new(12,node8.Next); [2]; node8.Next = inserted; [1]; 8→12→20→30 [2].

Q3. [6] Set node10.Next=node10.Next!.Next [2] to bypass 20 [1], leaving 10→30 [1]; finding node10 from head may take O(n) [1], but changing the link once identified is O(1) [1].

### Exit Ticket · Model answer

False. Insertion changes references; later node values do not need shifting.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**

**BLACK PEN** Answer all questions first.

**GREEN PEN** Then: tick correct answers; rewrite mistakes.

**Do Now** Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check my understanding